Package 'modelsirt'

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`modelsirt` *Explore the response curves of dichotomous, polytomous and multidimensional IRT models*

Description

Launches a Shiny app that draws the item response functions of the main item response theory (IRT) models in interactive plots, so that you can see how each item parameter shapes them: the probability of each possible response is plotted against the latent trait and redrawn as soon as a parameter is changed. The app covers the Rasch model and the one- to four-parameter logistic models for dichotomous items; the partial credit, generalized partial credit, graded response, rating scale, generalized rating scale and nominal response models for polytomous items; the probit response functions of Thurstonian IRT; and the response surface, contour plot and information surface of a dichotomous item that measures two abilities (multidimensional IRT).

Usage

```
modelsirt()
```

Details

Each tab plots, for a single item, the probability of each possible response as a function of the person's position on the latent trait (ability), written θ (η in the TIRT tab). The basic building block is the four-parameter logistic function

$$P(X = 1 \mid \theta) = c + \frac{d - c}{1 + \exp(-Da(\theta - b))}$$

the probability of a correct (or keyed) response, $X = 1$, where a is the discrimination (how steeply the probability rises with ability), b the difficulty (the ability at which the probability is halfway between its lower and upper limits), c the lower asymptote (the probability at very low abilities, as with guessing) and d the upper asymptote (the probability at very high abilities). D is the scaling constant: 1 keeps the logistic metric, and 1.702 makes the logistic curve almost coincide with the normal ogive, the cumulative normal distribution. The Rasch-4PL tab plots this function itself, the GRM tab uses it with $c = 0$ and $d = 1$ for the boundaries between categories, and the MIRT Dichotomous tab extends it to two abilities; the other models are given in their sections.

The five tabs on a single trait, Rasch-4PL to NRM, share a layout. A Scaling Constant choice sets D to 1 (the default) or 1.702. The ability axis runs from -6 to 6, so a curve centred beyond that range, which most location sliders allow (they reach -10 and 10), moves out of view. The plots are interactive plotly charts with a marker at every 0.1 of ability: hovering shows the values, clicking a legend entry hides or shows its curve, and the plot can be zoomed and panned. A horizontal gray line marks a probability of 0.5, and the vertical line at 0 is the zero line of the axis. Every plot is redrawn as soon as a control changes.

The app has seven tabs, described in the sections below.

Value

A Shiny app object (`shiny::shinyApp()`). Printing it, which is what happens when you call `modelsirt()` at the console, starts the app in the browser or the RStudio viewer; you can also pass it to `shiny::runApp()`.

Rasch-4PL

The item characteristic curve of a dichotomous item under the four-parameter logistic model (4PL) and the models nested in it, which fix some of its parameters.

- **Controls:** the scaling constant; the discrimination a (-10 to 10, default 1); the difficulty b (-10 to 10, default 0); the guessing parameter c (0 to 1, default 0); the inattentiveness parameter d (0 to 1, default 1); and the probability to plot: $P(X = 1 \mid \theta)$ alone, or $P(X \mid \theta)$ (the default), which adds the probability of an incorrect response, $P(X = 0 \mid \theta) = 1 - P(X = 1 \mid \theta)$.
- **Plot:** the chosen curves across ability, with a legend. The title names the model that the current values define: Rasch Model when $a = 1$, $c = 0$ and $d = 1$; 1PL or 2PL Model when $c = 0$ and $d = 1$ but a is not 1 (with a single item, a discrimination shared by all items, as in the 1PL, cannot be told apart from one specific to the item, as in the 2PL); 3PL Model when $c > 0$ and $d = 1$; 4PL Model (c = 0) when $c = 0$ and $d < 1$, a 4PL without guessing whose

upper asymptote allows for slipping (careless errors by able respondents); and 4PL Model when $c > 0$ and $d < 1$.

- **Text:** below the controls, the name of the model followed by the values of a, b, c and d.

Raising b moves the curve to the right, so that a higher ability is needed for the same probability. The discrimination sets the steepness: at $\theta = b$ the slope is $Da(d - c)/4$, a negative a makes the probability fall as ability rises, and $a = 0$ makes the curve flat. Raising c lifts the lower end of the curve, so that even the least able answer correctly with probability c, as when guessing; lowering d pulls the upper end down, so that even the most able miss with probability $1 - d$, as through inattention. With $c = 0$ and $d = 1$ the curve crosses the gray line at $\theta = b$, where the curves of $P(X = 1 \mid \theta)$ and $P(X = 0 \mid \theta)$ also cross; in general it reaches the midpoint $(c + d)/2$ there. Choosing $D = 1.702$ makes the curve steeper by that factor.

PCM-GPCM

Category response curves of an item scored in five ordered categories, 1 to 5, under the partial credit model (PCM) and the generalized partial credit model (GPCM):

$$P(X = k \mid \theta) = \frac{\exp\left(\sum_{v=1}^{k-1} Da(\theta - \delta_v)\right)}{\sum_{h=1}^{5} \exp\left(\sum_{v=1}^{h-1} Da(\theta - \delta_v)\right)}$$

for $k = 1, \ldots, 5$, where a is the discrimination, $\delta_1, \ldots, \delta_4$ are the step parameters (one for each step from a category to the next) and an empty sum is 0, so that category 1 has the numerator 1.

- **Controls:** the scaling constant; the discrimination a (-10 to 10 in steps of 0.5, default 1); and the four step parameters (-10 to 10), with defaults -2, -0.7, 0.8 and 2 (the code asks for -0.75 and 0.75, which the sliders round to one decimal).
- **Plot:** the probabilities of the five categories, $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each. The title reads Partial Credit Model (Masters, 1982) when $a = 1$ and Generalized Partial Credit Model (Muraki, 1992) otherwise. A discrimination of 0 is rejected with an error message in place of the plot.

The curves of adjacent categories k and $k+1$ cross exactly at $\theta = \delta_k$, so each step parameter moves one crossing point. With the step parameters in increasing order, each category in turn is the most likely response as ability rises; if a step parameter is lower than the one before it (reversed steps), the category between them is never the most likely. A larger a makes the curves narrower and more peaked, and a negative a reverses the order of the categories along the ability axis.

GRM

Category response curves of an item with five ordered categories under the graded response model (GRM). The model first gives, for each of the four boundaries between adjacent categories, the probability of a response in a category above the boundary,

$$P_k^*(\theta) = \frac{1}{1 + \exp(-Da(\theta - \delta_k))}, \quad k = 1, \ldots, 4,$$

and then the probability of each category as the difference between successive boundary curves, $P(X = m \mid \theta) = P_{m-1}^*(\theta) - P_m^*(\theta)$ for $m = 1, \ldots, 5$, with $P_0^* = 1$ and $P_5^* = 0$.

- **Controls:** the scaling constant; the discrimination a (0 to 10, default 1.5), shared by all boundaries; and the four difficulties, δ_1 (-2.5 to -1.5), δ_2 (-1.25 to -0.25), δ_3 (0 to 1.25) and δ_4 (1.5 to 2.5), with defaults -2, -0.8, 0.8 and 2 (the code asks for -0.75 and 0.75, which the sliders round to one decimal). The slider ranges do not overlap, which keeps the difficulties in increasing order, as the model requires: otherwise some category probabilities would be negative. For the same reason the discrimination cannot be negative: the boundary curves would then fall instead of rise and swap order, so an item that runs the other way is reverse-scored instead. At $a = 0$ the item does not depend on ability: every boundary curve is flat at 0.5, so categories 1 and 5 each get probability 0.5 and the middle categories none.
- **Plot:** $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each, titled Graded Response Model.

Each difficulty δ_k is the ability at which a response above category k becomes as likely as not. Only the first and the last can be read directly from the plot: the curve of category 1 crosses the gray line at δ_1, and that of category 5 at δ_4. Unlike the step parameters of the PCM, the difficulties are not the points where adjacent curves cross. Because the GRM gives the item a single discrimination, changing a makes all curves steeper or flatter together.

RSM-GRSM

Category response curves of an item with five ordered categories under the rating scale model (RSM) and its generalized form with a discrimination parameter (GRSM). Both are partial credit models in which each step parameter is split into the difficulty of the item and a threshold:

$$P(X = k \mid \theta) = \frac{\exp\left(\sum_{v=1}^{k-1} Da(\theta - (\delta + \tau_v))\right)}{\sum_{h=1}^{5} \exp\left(\sum_{v=1}^{h-1} Da(\theta - (\delta + \tau_v))\right)}$$

where δ is the difficulty (location) of the item and $\tau_1, \ldots, \tau_4$ are the thresholds, which in a rating scale model are the same for all items that share a response scale.

- **Controls:** the scaling constant; the discrimination a (-10 to 10, default 1); the difficulty δ (-10 to 10, default 0); and the four thresholds (-10 to 10; defaults -1, -0.4, 0.4 and 1).
- **Plot:** $P(X = 1 \mid \theta)$ to $P(X = 5 \mid \theta)$, one coloured curve each. The title reads Rating Scale Model (Andrich, 1978) when $a = 1$ and Generalized Rating Scale Model (Muraki, 1990) otherwise. A discrimination of 0 is rejected with an error message in place of the plot.

Adjacent curves k and $k + 1$ cross at $\theta = \delta + \tau_k$. Changing the difficulty shifts all five curves along the ability axis together without changing their shape, the thresholds set the spacing of the crossings around the difficulty, and the discrimination sets how peaked the curves are.

NRM

Category response curves of an item with five unordered categories, labelled a to e, under the nominal response model (NRM). Category a is the reference category, whose exponent is fixed at 0, and each of the other four categories has its own discrimination α_k (written with a Greek letter here, to avoid confusion with category a) and difficulty δ_k:

$$P(X = k \mid \theta) = \frac{\exp(z_k)}{\sum_{h=a}^{e} \exp(z_h)}, \quad z_a = 0, \quad z_k = D\alpha_k(\theta - \delta_k) \quad (k = b, \ldots, e).$$

Following Baker (1992), each exponent is written with a difficulty rather than an intercept.

- **Controls:** the scaling constant; four discriminations (-10 to 10; defaults -4, -2, 2 and 4) and four difficulties (-10 to 10; defaults -4, -2, 2 and 4). Discrimination j and difficulty j belong to the j-th category after a: 1 to b, 2 to c, 3 to d and 4 to e.
- **Plot:** $P(X = a \mid \theta)$ to $P(X = e \mid \theta)$, one coloured curve each, titled Nominal Response Model, with a note on a second, smaller line of the title that Baker's (1992) reparameterization is used.

Category k is exactly as likely as category a at $\theta = \delta_k$, because the ratio of their probabilities is $\exp(z_k)$. The discriminations order the categories along the ability axis: the category with the largest discrimination (counting 0 for category a) becomes the most likely at high abilities, the one with the smallest at low abilities, and the others can be the most likely in between. Nothing forces an order on the categories, which is what sets the NRM apart from the models for ordered categories. With the default values, categories c, a and d are in turn the most likely across the plotted range, while b and e would take over only beyond -6 and 6.

TIRT

Probit response functions of Thurstonian item response theory (TIRT), in which a binary outcome, such as the choice between two items in a forced-choice questionnaire, depends on the latent trait η through Φ, the cumulative distribution function of the standard normal distribution. The tab has three sub-tabs, each named by a formula, with these functions and controls:

- **First sub-tab:** a single binary outcome, written with the parameters of factor analysis,

$$P(\eta) = \Phi\left(\frac{-\gamma + \lambda\eta}{\sqrt{\psi}}\right),$$

  with the threshold γ (-10 to 10, default 0), the factor loading λ (-10 to 10, default 0.5) and the error (unique) variance ψ (0 to 10, default 0.5).
- **Second sub-tab:** the choice of item i over item k in a pair,

$$P(\eta) = \Phi\left(\frac{-\gamma + \lambda_i\eta - \lambda_k\eta}{\sqrt{\psi_i + \psi_k}}\right),$$

  with the threshold of the pair γ (-10 to 10, default 0), the loadings of the two items λ_i and λ_k (-10 to 10; defaults 0.5 and 0) and their error variances ψ_i and ψ_k (0 to 10, default 0.5 each). The formula on the sub-tab gives each item its own trait, but the plot uses the same η for both, so only the difference $\lambda_i - \lambda_k$ matters: with equal loadings the curve is flat, and the choice tells nothing about the trait.
- **Third sub-tab:** the same comparison in intercept and slope form,

$$P(\eta) = \Phi(\alpha + (\beta_i - \beta_k)\eta),$$

  with the intercept α (-10 to 10, default 0) and the slopes β_i and β_k (-10 to 10; defaults 0.5 and 0). It draws the curve of the second sub-tab when $\alpha = -\gamma/\sqrt{\psi_i + \psi_k}$ and each $\beta = \lambda/\sqrt{\psi_i + \psi_k}$; with their defaults, both sub-tabs show $\Phi(0.5\eta)$.
- **Range:** each sub-tab also has a slider labelled η (1 to 100, default 6) that sets no parameter: it sets the range of the horizontal axis, from $-\eta$ to η.
- **Plot:** the probability, from 0 to 1, against η: a single curve with a marker at every 0.1, and the gray line at 0.5.

Each curve crosses 0.5 where the argument of Φ is 0, for example at $\eta = \gamma/\lambda$ in the first sub-tab. The loadings or slopes set the steepness and direction of the curve, and larger error variances make it flatter. There is no scaling constant in this tab, since the curves are normal ogives.

MIRT Dichotomous

The response surface of a dichotomous item that measures two abilities, θ_1 and θ_2 (multidimensional IRT, MIRT), on a grid of 50 by 50 points from -4 to 4 on each dimension. In the compensatory model (the default) the probability is a logistic function of a weighted sum of the two abilities,

$$P(\theta_1, \theta_2) = c + \frac{d - c}{1 + \exp(-D(a_1(\theta_1 - \delta_1) + a_2(\theta_2 - \delta_2)))},$$

so that a high ability on one dimension can make up for a low ability on the other. In the non-compensatory model it is the product of one logistic function for each ability,

$$P(\theta_1, \theta_2) = c + (d - c)\frac{1}{1 + \exp(-Da_1(\theta_1 - \delta_1))} \cdot \frac{1}{1 + \exp(-Da_2(\theta_2 - \delta_2))},$$

so that a low ability on either dimension keeps the probability low, whatever the other. Here a_1 and a_2 are the discriminations, δ_1 and δ_2 the locations on the two dimensions, and c and d the lower and upper asymptotes.

- **Controls:** a choice labelled Item Response Function, which sets the scaling constant of the logistic function: Logistic gives $D = 1$, and Normal ogive approximation (the default) gives $D = 1.702$, which makes the logistic curve almost identical to the normal ogive; whether the model is compensatory (Yes, the default, or No); the item's location on each dimension, labelled θ_1 location and θ_2 location (-10 to 10, default 0); its discrimination on each dimension, labelled θ_1 discrimination and θ_2 discrimination (-10 to 10, default 1; a value of 0 is rejected with an error message); the lower asymptote c (0 to 0.4 in steps of 0.1, default 0) and the upper asymptote d (0 to 1, default 1); and optional names for the two dimensions. The axes are labelled θ_1 and θ_2, and a name typed in is shown in front of the symbol, for example Math (θ_1).
- **Item Response Surface sub-tab:** $P(\theta_1, \theta_2)$ as an interactive 3-D surface, coloured from purple (low) to yellow (high probability), with P(Y = 1) from 0 to 1 on the vertical axis; drag it to rotate it.
- **Contour Plot sub-tab:** the same surface seen from above, as contour lines of equal probability, each labelled with its probability; a slider sets the approximate number of contour levels (10 to 100, default 10). Black lines mark $\theta_1 = 0$ and $\theta_2 = 0$. A thick line marks the contour halfway between the asymptotes, at probability $(c+d)/2$ (0.5 when $c = 0$ and $d = 1$); it is left out when that probability is not reached inside the grid, as for a very easy or very hard item. For the compensatory model a blue arrow shows the item vector. It starts on the line where the probability is halfway between the asymptotes, at the point of that line closest to the origin; it points in the direction in which the probability rises fastest, at the angle $\arccos(a_1/A)$ to the θ_1 axis; and its length is the multidimensional discrimination $A = \sqrt{a_1^2 + a_2^2}$. Its starting point lies at the signed distance $(a_1\delta_1 + a_2\delta_2)/A$, the multidimensional difficulty, from the origin along the direction of the arrow.
- **Information Plot sub-tab:** a 3-D surface, seen from the opposite corner, of the item information at each point of the grid: the squared slope of the probability surface, in the direction in

which it rises fastest, divided by $P(1-P)$,

$$I(\theta_1, \theta_2) = \frac{(\partial P/\partial\theta_1)^2 + (\partial P/\partial\theta_2)^2}{P(1-P)}.$$

For the compensatory model this is the information of the four-parameter logistic model (Magis, 2013) along the item vector,

$$I(\theta_1, \theta_2) = \frac{D^2A^2(P-c)^2(d-P)^2}{(d-c)^2P(1-P)},$$

which grows with the multidimensional discrimination A and, when $c = 0$ and $d = 1$, is highest along the line where $P = 0.5$. For the non-compensatory model the slope combines the two logistic parts of the model.

In the compensatory model the contour lines are straight and parallel, at right angles to the item vector, because the probability depends only on the weighted sum of the abilities; in the non-compensatory model they bend, since both abilities must be high for a high probability. Making one discrimination larger than the other turns the item vector towards the axis of that dimension, which means that the item measures mostly that ability, and moving a location shifts the surface along its dimension.

Acknowledgments

modelsirt is a fork and substantial extension of the irtDemo package by Wes Bonifay and Metin Bulus, which is available on CRAN (https://cran.r-project.org/package=irtDemo); the title of the app, IRT Models using IRTDemo, reflects this origin. Dimitrios Zacharatos modified and expanded it: he consolidated the IRT models into a single interface, rebuilt the visualisations with Plotly for interactivity, added further IRT models and improved the user interface. Special thanks to Kostas Maistrelis (https://github.com/maisk) for contributing the implementation of the Thurstonian IRT curves. The tabs keep the credits shown in the app itself: the Rasch-4PL, PCM-GPCM and RSM-GRSM tabs are credited to Metin Bulus, and the GRM, NRM and MIRT Dichotomous tabs to an unknown contributor.

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Examples

```
if (interactive()) modelsirt()
```

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